

1. Show that the sliced score matching (SSM) loss can also be written as

$$L_{SSM} = \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [\|v^T S(x; \theta)\|^2 + 2v^T \nabla_x (v^T S(x; \theta))].$$

Define 單-投影損失 $J_v(\theta) := \mathbb{E}_{x \sim p(x)} [(v^T S(x; \theta) - v^T \nabla_x \log p(x))^2]$

then SSM's total loss

$$\begin{aligned} L_{SSM} - \text{intractable} &= \mathbb{E}_{v \sim p(v)} [J_v(\theta)] \\ &= \mathbb{E}_v \mathbb{E}_x [(v^T S(x; \theta) - v^T \nabla_x \log p(x))^2] \\ &= \mathbb{E}_v \mathbb{E}_x [\underbrace{(v^T S(x; \theta))^2}_A - \underbrace{2(v^T S(x; \theta))(v^T \nabla_x \log p(x))}_B + \underbrace{(v^T \nabla_x \log p(x))^2}_C] \end{aligned}$$

A: depend on $S(x; \theta)$

B: depend on $S(x; \theta)$ and $\nabla_x \log p(x)$

C: only depend on $p(x)$. 忽略

$\Rightarrow$ 需优化的 tractable Loss $L_{SSM} \propto \mathbb{E}_v \mathbb{E}_x [(v^T S(x; \theta))^2 - 2(v^T S(x; \theta))(v^T \nabla_x \log p(x))] - (\Delta)$

Let $I = \mathbb{E}_x [(v^T S(x; \theta))(v^T \nabla_x \log p(x))]$

$$= \int_{\mathbb{R}^d} p(x) [(v^T S(x; \theta))(v^T \nabla_x \log p(x))] dx \quad (\text{by Divergence Thm})$$

and by the def of score function $\nabla_x \log p(x) = \frac{\nabla_x p(x)}{p(x)}$, we have

$$I = \int_{\mathbb{R}^d} p(x) (v^T S(x; \theta)) (v^T \frac{\nabla_x p(x)}{p(x)}) dx$$

$$\text{Thus } I = \int_{\mathbb{R}^d} (v^T S(x; \theta)) (v^T \nabla_x p(x)) dx$$

Let $g(x) = v^T S(x; \theta)$, $g: \mathbb{R}^d \rightarrow \mathbb{R}$

v is constant vector,

$\nabla_x p(x)$ is vector field

and $v^T \nabla_x p(x)$ 是 $p(x)$ 在 v 方向上的 directional derivative, as $v \cdot \nabla p(x)$

$$\text{then } I = \int_{\mathbb{R}^d} g(x) (v \cdot \nabla p(x)) dx$$

By Green's first identity: $\int_{\Omega} u(\nabla \cdot F) dV = \int_{\partial \Omega} u(F \cdot \hat{n}) dS - \int_{\Omega} (\nabla u \cdot F) dV$

let $u = g(x)$, $F = v \cdot p(x)$.

$$\text{Then } I = \int_{\mathbb{R}^d} g(x) (\nabla \cdot F) dx = \underbrace{\int_{\partial \Omega} g(x) (F \cdot \hat{n}) dS}_{(*)} - \underbrace{\int_{\Omega} (\nabla g(x) \cdot F) dV}_{(**)}$$

For $(*)$ $\int_{\partial \Omega} (v^T S(x; \theta)) (v \cdot p(x) \cdot \hat{n}) dS$ 在 x 的無窮遠 $\partial \Omega$ 上,

In Score matching, we suppose that when $x \rightarrow \infty$, $p(x) \rightarrow 0$ 的速度夠快 such that $p(x) S(x; \theta) \rightarrow 0$

Thus, $(*) = 0$.

Thus, we only need to deal with $(**) = -\int_{\Omega} (\nabla g(x) \cdot F) dx$, 代回 $g(x)$ and F , we have

$$\begin{aligned} I &= -\int_{\Omega} (\nabla_x [u^T S(x; \theta)] \cdot [v \cdot p(x)]) dx, \\ &= -\int_{\Omega} p(x) (\nabla_x [u^T S(x; \theta)] \cdot v) dx \\ &= -\int_{\Omega} p(x) (u^T \nabla_x [u^T S(x; \theta)]) dx \\ &= -E_{x \sim p(x)} [u^T \nabla_x (u^T S(x; \theta))] = (66) \end{aligned}$$

By (6), $L_{SSM} \propto E_u E_x [(u^T S(x; \theta))^2 - \underbrace{2(u^T S(x; \theta))(u^T \nabla_x \log p(x))}_I]$

By (66), we have

$$\begin{aligned} L_{SSM} &= E_u E_x [(u^T S(x; \theta))^2 - 2(-u^T \nabla_x (u^T S(x; \theta)))] \\ &= E_u E_x [(u^T S(x; \theta))^2 + 2 u^T \nabla_x (u^T S(x; \theta))] \\ &= E_{x \sim p(x)} E_{u \sim p(u)} [\|u^T S(x; \theta)\|^2 + 2 u^T \nabla_x (u^T S(x; \theta))] \quad (\because u^T S(x; \theta) \text{ is scalar}) \end{aligned}$$

2. Briefly explain SDE.

• SDE 是一種用來描述系統狀態如何隨時間演變的微分方程，但和 ODE 不同，因為 SDE 的演化過程包含 stochastic and noise.

• SDE 的一般形式： $dx = f(x, t) dt + g(x, t) dw$
where $f(x, t) dt$ is Drift, $g(x, t) dw$ is Diffusion.

• SDE 在 Score-Based Models 中的應用：

1. Forward Process : 將 real information $p_0(x) \xrightarrow{t}$ 純 noisy $p_t(x)$
2. Reverse Process : 1. 的反向： $p_t(x) \xrightarrow{t} p_0(x)$
3. 核心零件： $\nabla_x \log p_t(x)$ (Score function)

summary: SDE 提供了「加 noise $\leftrightarrow$ 生成」的連續框架。

目標：train 神經網路 $S(x, t; \theta)$ 來估計 Score $\nabla_x \log p_t(x)$,

再將 S 代入反向 SDE 中，就能從噪聲還原出新的數據

3. Unanswered Questions

There are unanswered questions from the lecture, and there are likely more questions we haven't covered.

3. Euler-Maruyama 中，為什麼 drift 乘以 Δt ，但 diffusion 是乘以 $\sqrt{\Delta t}$?